MINH NHAT NGUYEN

AI Engineer | LLM Post-Training | Agents | ML Systems

Ottawa, ON, Canada | nguyenminhnhatdtu@gmail.com | Authorized to work in Canada
Live demo: local-find-vibe17.vercel.app | github.com/Practicinginhell | linkedin.com/in/livetoevolve

PROFESSIONAL SUMMARY

AI engineer with 4+ years of experience, focused on LLM post-training, retrieval, and agent systems. Built a fine-tuning framework covering 5 training methods on GPU, Apple Silicon, and Google TPU, cutting held-out loss 38.9% on a 3B model. Led a 10-engineer team whose redesign of a live retrieval system raised accuracy 27%. Built Gleaner solo for the Meta Global AI Developer Hackathon, a local-first research assistant on Meta's Llama and Muse models with 423 tests. Leads a 5-person AI team in a 22-person healthcare research programme while in a graduate AI & ML program (GPA: 3.86).

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, Java, C#, C++, Swift, SQL, Cypher
- **LLMs & Agents:** RAG, tool calling, MCP, LangGraph, LangChain, CrewAI, Gemini API, OpenAI API, Meta Model API, Ollama, vLLM
- **Training:** PyTorch, JAX, MLX, Hugging Face TRL and PEFT, LoRA, QLoRA, SFT, DPO, KTO, ORPO, GRPO, Google TPU, CUDA
- **Engineering:** FastAPI, Next.js, React Native, Docker, GitHub Actions CI, pytest, Sentry, Git, Linux
- **Data:** PostgreSQL, MySQL, SQLite, Neo4j, Databricks, vector databases, MinHash deduplication, forecasting (TimesFM)

KEY PROJECTS

Gleaner: AI Research Assistant on Meta Models

Oct 2026 - Nov 2026

Meta Global AI Developer Hackathon (Cerebral Valley), github.com/Practicinginhell/gleaner — Solo, Indie track

Python · Llama 3.2 · Muse Spark · Ollama · Graphflow · FastAPI · SQLite

- Built a local-first app in 9 days that reads arXiv, Hacker News, GitHub releases, and blogs, and keeps a user's standing questions answered with word-for-word, linked quotes.
- Cut 1,305 collected items to 17 confirmed quotes with a filter cascade of rules, MinHash deduplication, an embedding gate, a prompt-injection screen, and a local Llama 3.2 judge.
- Measured small models on real items and designed around the results: the 3B model accepted a sentence under the wrong question 1 time in 21, versus 21 of 21 for 1B, so models answer only yes/no and code does the rest.
- Added Muse Spark cited answers, web search for new feeds, voice control at about 1.5 s per command, and an MCP server for coding agents, backed by 423 offline tests.

Graphflow: Agentic AI Framework

Personal Project

Python · LangGraph · LangChain · Pydantic

- Created a framework for building modular agents and multi-step automations in a few lines of code, with typed tools, structured output, and human approval before sensitive actions.
- Hardened agents with retries, timeouts, fallbacks, and saved history, verified by 101 tests that run offline.

AgentOps: Reliability and Observability Runtime for AI Agents

Oct 2026

Hackathon project, github.com/CommunityHackathons/adai26-team-008 — Team project

Python · Graphflow (LangGraph) · FastAPI · SQLite · Ollama · Gemini

- Built a runtime that wraps coding agents in a deterministic state machine: the agent patches code in an isolated sandbox, pytest verifies the fix, and a failure triggers one bounded revision on a stronger model.
- Routed read-only triage to a fast model tier at about 8x lower cost, and guarded every tool call with path-traversal rejection and blocked edits to tests.
- Logged every step, diff, token count, and cost to SQLite with live streaming, and benchmarked pass rate and cost per verified fix against a single-pass baseline, backed by 33 tests.

TRL Pipeline: LLM Training and Evaluation Framework Personal Project

Python · PyTorch · JAX · Hugging Face TRL · Google TPU · GitHub Actions

- Designed a pipeline where one config file runs data preparation, fine-tuning, evaluation, and publishing across 5 training methods, trained on Google TPU v5e-8 through a Google Research program.
- Engineered a reinforcement learning environment for C++ code generation that runs model output against test cases under resource limits and rewards the fraction of tests passed.
- Prepared a 15,655-example dataset from 1.18M raw rows, and audited it to find 24% low-quality records that automated checks had passed.
- Cut held-out loss 38.9% in controlled comparisons, and showed a lighter method matched full fine-tuning at ~8% of the parameters, backed by 525 tests.

LocalFind: AI-Powered Inventory Search Platform Jun 2026 - Aug 2026

Solo Developer, live at local-find-vibe17.vercel.app Ottawa, ON

Python · FastAPI · PostgreSQL · React · Gemini API · Built with Claude Code

- Took an idea to a live product in 3 months, with 108 API endpoints helping shoppers find in-stock products nearby.
- Built an AI assistant with Gemini function calling, plus an on-device model sharing the same tools so it works offline, verified by 172 automated tests.

TankLight: Water Safety Tracker and Delivery Planner Sep 2026

Hack for Humanity Ottawa 2026, github.com/Practicinginhell/tank_light Ottawa, ON

Python · TimesFM · Ollama (Gemma 3) · Graphflow

- Forecast urgent deliveries with TimesFM against a simple baseline on a simulated 12-home town, and cut hours homes went without water from 534 to 195 in July with the same number of stops.

PROFESSIONAL EXPERIENCE

Healthcare AI Research Collaboration Remote, Canada / Denmark

AI Team Lead (5 engineers, 22-person programme) 2025 - Present

- Leads a 5-person AI team within a 22-person international programme directed by a professor from Denmark, applying machine learning to healthcare problems.
- Turns open-ended research goals into scoped tasks, runs technical discussions, and presents results to clinical, non-technical collaborators.

Ftech Company Da Nang, Vietnam

AI Researcher & Team Manager Aug 2021 - Jul 2024

- Led a 10-engineer team that redesigned a live RAG system with hybrid search and reranking, measuring a 27% accuracy gain against the previous system.
- Automated a multi-source data preparation pipeline for LLM fine-tuning, cutting manual effort by ~60% and turnaround from weeks to days.
- Built a question-answering system over a Neo4j knowledge graph that turned plain-language questions into Cypher queries, beating the keyword-search baseline.
- Delivered AI agent workflows, authored a Flax/NNX training framework used for in-house experiments, and managed a 4-engineer team through sprint planning and code reviews.

EDUCATION

Lambton College Ottawa, ON, Canada

Artificial Intelligence & Machine Learning (AIMO), currently enrolled, GPA: 3.86/4.0 2025 - Present

Duy Tan University Da Nang, Vietnam

B.IT: Software Engineering (CMU Program) 2014 - 2018

ADDITIONAL INFORMATION

- **Research Community:** Current member of Cohere Labs, the open science research community of Cohere
- **Certifications:** UC Berkeley Agentic AI Professional Certificate (MOOC), IT Passport (IPA Japan)
- **Languages:** English (Fluent), Vietnamese (Native)